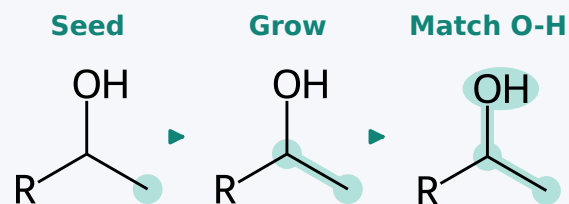
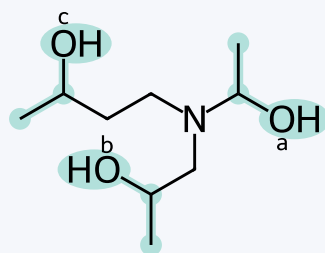


a GRAFT growth: earlier placements condition every later choice

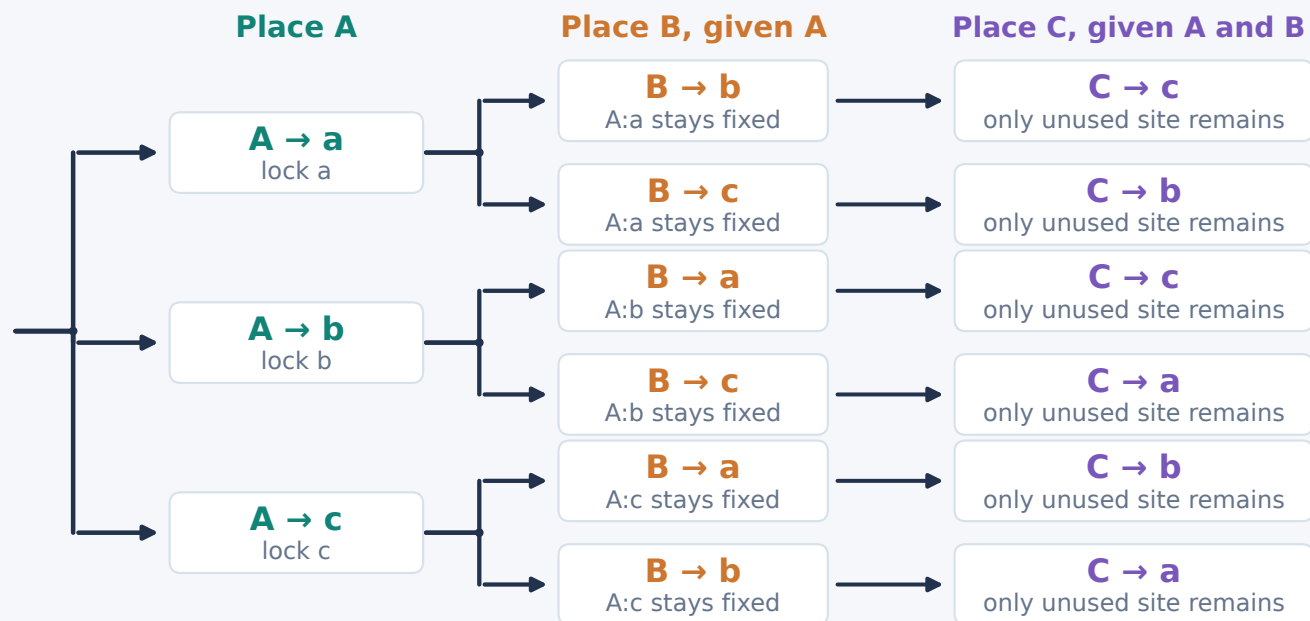
Local alcohol fragments A, B and C share this motif; R lies outside the matched atoms.



Three distinct target sites



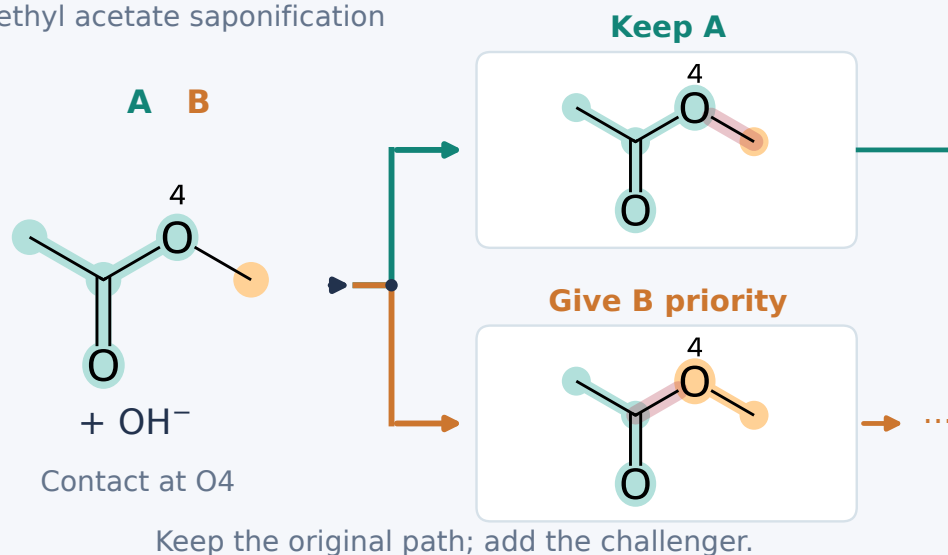
Each site preserves C-C, C-O and O-H.



Each path is a joint assignment; occupied target atoms cannot be reused.

b Add a branch when fragments compete

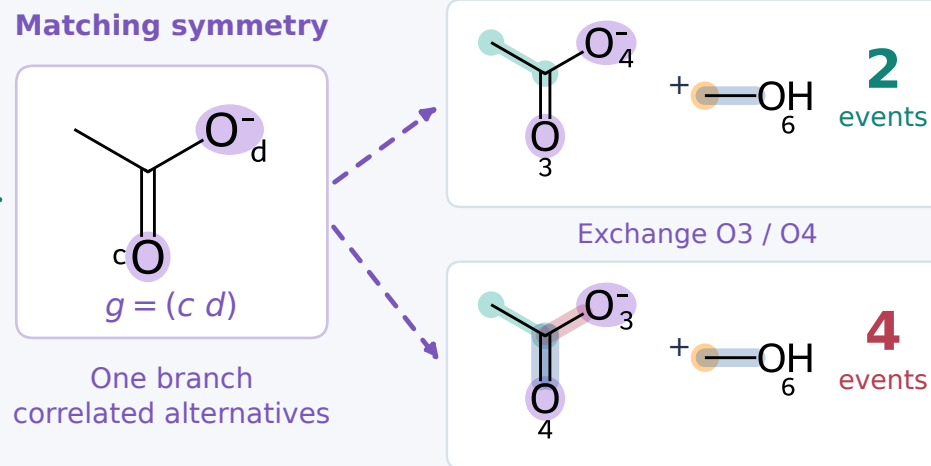
Methyl acetate saponification



Keep the original path; add the challenger.

c Decode each compressed family

The oxygen exchange preserves matching at $\tau_{iso} = 1$



→ Search continuation

- - - Event projection

Red: lose/weaken · blue: form/strengthen